

SWE 4633

Module 4

Cloud Software Development

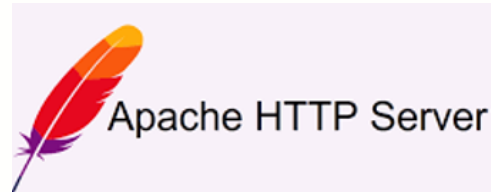
One Example of a Cloud-Base Web Application

- EC2: virtual machine and a server
 - Connect with S3 bucket and RDS
 - Development environment
- S3: store large files or images
 - The location of each file in S3 can also be stored in RDS as one attribute



Web Server

- Installed in the EC2 instances
- The basic job is to accept requests from clients (e.g., a visitor's web browser) and then send the response to the request, e.g.:
 - Display the page that a visitor wants to see
 - Store user inputs to back-end database
 - Retrieve data from back-end database
 - ...
- Apache web server: A popular open-source, cross-platform web server



Install Apache Web Server in EC2

- Fetch the latest version of the package that can be installed on the system

sudo apt-get update

- Install Apache web server

sudo apt-get install apache2

- Start Apache server

sudo service apache2 status

Install Apache Web Server in EC2

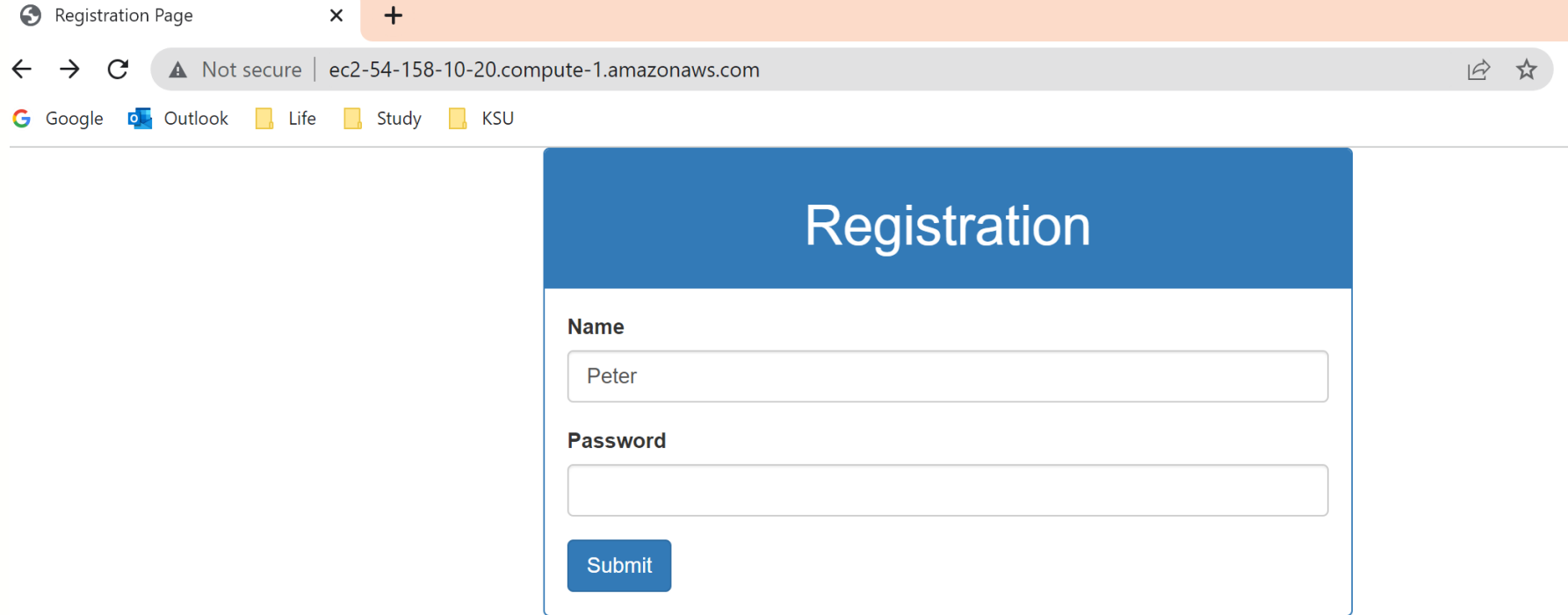
- The application code should be stored in */var/www/html*

```
*** System restart required ***  
Last login: Tue Sep  5 19:38:42 2023 from 168.28.186.189  
ubuntu@ip-172-31-38-211:~$ cd /var/www/html/  
ubuntu@ip-172-31-38-211:/var/www/html$ ls  
connect.php  css  index.html
```

- Change permission of the folder so that we can edit it
sudo chmod 777 /var/www/html
- The webpage under this folder can be displayed by typing the URL of the EC2 instance
 - e.g., “ec2-54-158-10-20.compute-1.amazonaws.com”
 - Open this URL address in any browser



Example 1



The image is a screenshot of a web browser window. The browser's address bar shows the URL "ec2-54-158-10-20.compute-1.amazonaws.com" with a "Not secure" warning. The browser's tab bar shows a single tab titled "Registration Page". The browser's search bar shows "Google". The browser's bookmarks bar shows "Outlook", "Life", "Study", and "KSU". The main content of the browser is a registration form. The form has a blue header with the word "Registration" in white. Below the header, there are two input fields: "Name" and "Password". The "Name" field contains the text "Peter". Below the "Password" field is a blue "Submit" button.

Registration Page x +

← → ↻ ⚠ Not secure | ec2-54-158-10-20.compute-1.amazonaws.com 🔗 ☆

🔍 Google 📧 Outlook 📁 Life 📁 Study 📁 KSU

Registration

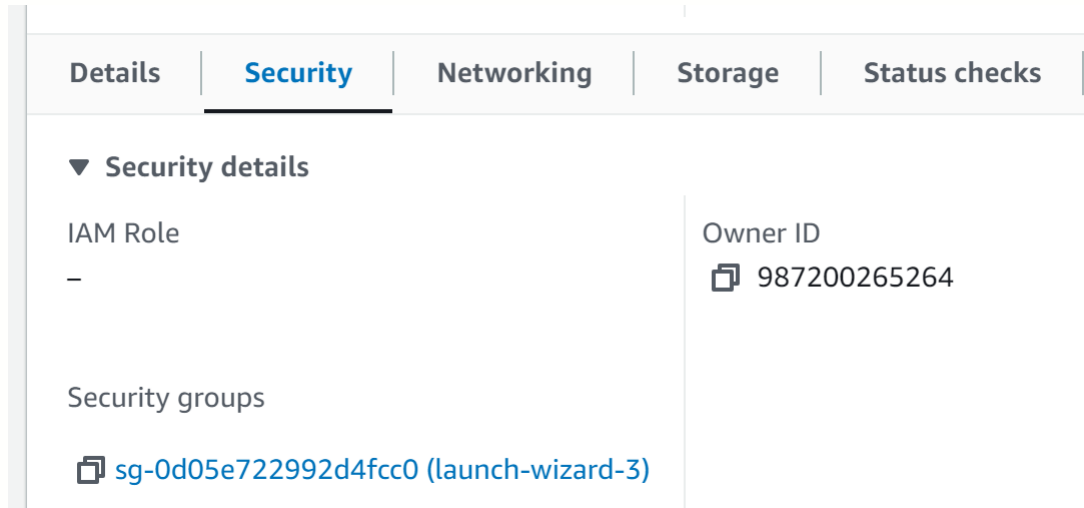
Name

Password

Submit

Set Inbound Rules

- If the webpage can't be displayed, the possible reason is that the *Http or Https* is not allowed in inbound rules of security group.
- Add inbound rules as the following pages



The screenshot shows the AWS IAM console interface. At the top, there are tabs for 'Details', 'Security' (which is selected and underlined), 'Networking', 'Storage', and 'Status checks'. Below the tabs, the 'Security details' section is expanded, showing 'IAM Role' with a value of '-' and 'Owner ID' with a value of '987200265264'. Under the 'Security groups' section, there is a link to 'sg-0d05e722992d4fcc0 (launch-wizard-3)'.

Details	Security	Networking	Storage	Status checks
▼ Security details				
IAM Role		Owner ID		
-		987200265264		
Security groups				
sg-0d05e722992d4fcc0 (launch-wizard-3)				

Edit inbound rules [Info](#)

Inbound rules control the incoming traffic that's allowed to reach the instance.

Inbound rules [Info](#)

Security group rule ID	Type Info	Protocol Info	Port range Info	Source Info	Description - optional Info	
sgr-0c1c1cf4a2a19a45c	HTTP ▼	TCP	80	Custom ▼ Q 0.0.0.0/0 ✕		Delete
sgr-0615c871e6eb57cc3	MYSQL/Aurora ▼	TCP	3306	Custom ▼ Q 0.0.0.0/0 ✕		Delete
sgr-04f49acf51dd5a226	HTTPS ▼	TCP	443	Custom ▼ Q 0.0.0.0/0 ✕		Delete
sgr-02812c154e91297ab	SSH ▼	TCP	22	Custom ▼ Q 0.0.0.0/0 ✕		Delete

Server-Side Language

- There are many server-side languages that are used to process information received from services, such as **PHP**, Python, etc.
- We will use PHP and *Form* in HTML to show how to connect the EC2 instance and AWS RDS database



```
<form action="connect.php" method="post">

  <div class="form-group">
    <label>Name</label>
    <input
      type="text"
      class="form-control"
      name="name"
    />
  </div>

  <div class="form-group">
    <label>Password</label>
    <input
      type="password"
      class="form-control"
      name="password"
    />
  </div>

  <input type="submit" class="btn btn-primary" />
</form>
```

Example of Sender and Receiver

Sender

Receiver

```
<?php
    $name = $_POST['name'];
    $password = $_POST['password'];
?>
```

- Receiver code is from “connect.php” specified in sender
- Information is retrieved via `$_POST[?]` where ? represent different labels

```
<form action="connect.php" method="post">

    <div class="form-group">
        <label>Name</label>
        <input
            type="text"
            class="form-control"
            name="name"
        />
    </div>

    <div class="form-group">
        <label>Password</label>
        <input
            type="password"
            class="form-control"
            name="password"
        />
    </div>

    <input type="submit" class="btn btn-primary" />
```

PHP

- Create a file with the extension .php and enclose any code with `<?php .... ?>`
- Use *action* in HTML to link the .php file
- Most cloud providers automatically recognize PHP with no additional configurations needed
- Languages like Python require a more set-up, which may vary from cloud platforms
 - <https://aws.amazon.com/developer/language/python/>
- Install PHP in EC2

sudo apt-get install php php-mysql

Connecting to the RDS

- We also need to add the inbound rules of RDS security group for *Mysql/aurora*

swe4633db Modify Actions ▼

Summary

DB identifier swe4633db	CPU 2.00%	Status 🟢 Available	Class db.t3.micro
Role Instance	Current activity 0 Connections	Engine MySQL Community	Region & AZ us-east-1c

Connectivity & security | Monitoring | Logs & events | Configuration | Maintenance & backups | Tags

Connectivity & security

Endpoint & port Endpoint swe4633db.cdxay7jffxd0.us-east-1.rds.amazonaws.com	Networking Availability Zone us-east-1c VPC	Security VPC security groups SWE4633group1 (sg-097ba41a8290d0727) 🟢 Active
--	---	--

Edit inbound rules [Info](#)

Inbound rules control the incoming traffic that's allowed to reach the instance.

Inbound rules [Info](#)

Security group rule ID	Type Info	Protocol Info	Port range Info	Source Info	Description - optional Info	
sgr-0b99f011a4dff3075	HTTP	TCP	80	Custom		Delete
				0.0.0.0/0		
sgr-00b11f3fda0eaca3b	MYSQL/Aurora	TCP	3306	Custom		Delete
				0.0.0.0/0		

Add rule

Setting of RDS

- Set the RDS instance NOT public access for the purpose of privacy.
- Instead, we set the security group of the RDS instance same with EC2 instance.

i-07d1fd2f08c05e6cc (webec2)

▼ Security details

IAM Role

—

Security groups

 sg-0d711098fed03b6c7 (launch-wizard-1)

Connectivity & security

Endpoint & port

Endpoint

 database-1.cf4u2ww2s
0c2.us-east-1.rds.amazonaws.com

Port
3306

Networking

Availability Zone

us-east-1c

VPC

vpc-00e55f96cd492a35b

Subnet group

default-vpc-
00e55f96cd492a35b

Security

VPC security groups

launch-wizard-1 (sg-
0d711098fed03b6c7)

✓ Active

~~swe4633db (sg-~~
066f338b13426d02e)

✓ Active

Publicly accessible

No

Connecting with RDS

- Since we set RDS as NOT public access, we can't access it remotely (via MySQL Workbench)
- We can enter EC2 to manage the database in command line

```
ubuntu@ip-172-31-89-231:~$ sudo mysql -u admin -p12345678 -h database-1.cf4u2ww2s0c2.us-east-1.rds.amazonaws.com
mysql: [Warning] Using a password on the command line interface can be insecure.
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 197
Server version: 8.0.35 Source distribution

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affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql>
```

Connecting to RDS

```
mysql> show databases;
```

```
+-----+  
| Database |  
+-----+  
| information_schema |  
| mysql |  
| performance_schema |  
| swe4633test |  
| sys |  
| webswe4633 |  
+-----+  
6 rows in set (0.00 sec)
```

```
mysql> CREATE TABLE webswe4633.registration (  
->     user_name varchar(255),  
->     user_password varchar(255)  
->  
-> );
```

```
mysql> select * from webswe4633.registration;
```

```
+-----+-----+  
| user_name | user_password |  
+-----+-----+  
| Harry    | 12345        |  
+-----+-----+  
1 row in set (0.00 sec)
```


Exercise

- Based on Example 2, create a table called “employee” with 4 columns: id, first_name, last_name and title
- Insert some records into the table
- Search the table by different names